

Engineering Architecture Standards

Principles

Services must be stateless where possible, expose health and readiness probes, and emit structured JSON logs. All inter-service communication uses authenticated HTTPS.

Data Layer

Each service owns its database. Cross-service data access happens only through published APIs or an event stream, never by reaching into another service's database directly.

API Standards

APIs follow REST conventions, are versioned under /v1, and must document every endpoint with OpenAPI. Breaking changes require a new major version and a deprecation window of 90 days.

Observability

Every service exports metrics, logs and traces. Service level objectives (SLOs) target 99.9% availability and a p99 latency under 300 ms for synchronous endpoints.